

Dead Man's Switch API

API Documentation v1.0

Overview

The Dead Man's Switch API provides a lightweight monitoring solution for critical systems. It implements a heartbeat-based health check mechanism that triggers alerts when systems fail to send periodic signals.

Base URL

```
http://localhost:8080
```

Content Type

All requests must include the following header:

```
Content-Type: application/json
```

API Endpoints

1. Register a Monitor

Request Body

Response: 201 Created — Monitor created successfully

Possible Errors:

2. Send Heartbeat

Parameter	Type	Description
id	String	Monitor ID

Response: 200 OK — Heartbeat received

Possible Errors:

3. Pause Monitor

Response: 200 OK — Monitor paused

Behavior

- Timer is stopped
- No alerts will be triggered
- Sending a heartbeat resumes monitoring

4. Get Monitor Status

Parameter	Type	Description
id	String	Monitor ID

```
{  "id": "device-123",  "timeout": 60,  "alertEmail": "admin@critmon.com",
  "paused": false,  "expiryTime": 1714900000000 }
```

Response: 200 OK — Monitor details returned

Possible Errors:

5. Delete Monitor

Parameter	Type	Description
id	String	Monitor ID

Response: 200 OK — Monitor deleted

Possible Errors:

Alert Behavior

If no heartbeat is received before the timeout period expires, an alert is automatically triggered.

Alert Payload

```
{  "ALERT": "Device device-123 is down!",  "time": "2026-05-05T12:00:00Z" }
```

Note: Alerts are logged via console. Monitor status transitions to DOWN.

Monitor States

State	Description
ACTIVE	Timer is running, monitoring is active
PAUSED	Timer is stopped, no alerts will trigger
DOWN	Timer expired, alert has been triggered

Error Response Format

All error responses follow this standardized format:

```
{  "error": "Monitor not found",  "status": 404,  "timestamp": "2026-05-05T12:00:00Z" }
```

Example Test Flow

- POST /monitors → Create a monitor
- POST /monitors/{id}/heartbeat → Send heartbeat
- POST /monitors/{id}/pause → Pause monitoring
- GET /monitors/{id} → Check status
- Wait (no heartbeat) → Alert triggered